

Digital Technology Adoption and Labor Productivity in Brazil:

An Analysis of the Period 2010–2023

1 INTRODUCTION

The relationship between technological change and economic productivity remains one of the most extensively debated topics in contemporary macroeconomics. Over the past two decades, the rapid diffusion of digital technologies — encompassing automation tools, artificial intelligence (AI), cloud computing, and digital platforms — has fundamentally transformed production processes across a wide range of industries worldwide. Despite substantial private investment in these technologies, however, many economies, particularly emerging markets, have failed to register commensurate gains in aggregate labor productivity. Brazil offers a compelling case in this regard: between 2010 and 2023, the country experienced a substantial expansion in the adoption of digital technologies, yet labor productivity growth remained persistently sluggish.

This apparent disconnect between technological investment and productivity outcomes is not without theoretical precedent. The so-called "productivity paradox" — the observation that periods of intense technological innovation do not necessarily translate into immediate or proportional gains in aggregate productivity — has long occupied scholars at the intersection of growth economics and the economics of innovation. The phenomenon was famously identified in the context of information-technology adoption in advanced economies and has gained renewed urgency in the era of AI and large-scale

automation. As Acemoglu and Restrepo (2018) argue in their task-based theoretical framework, automation technologies generate a powerful displacement effect by substituting machines for labor in tasks previously performed by workers. This displacement does not invariably produce net productivity gains unless it is accompanied by sufficiently strong countervailing forces, such as the creation of new labor-intensive tasks, capital accumulation, or meaningful productivity spillovers to non-automated sectors.

In the Brazilian context, such structural constraints appear particularly acute. The country's labor force is characterized by significant skill heterogeneity, marked regional disparities in technological infrastructure, and an institutional environment that has historically hindered the efficient reallocation of workers across sectors. These conditions are precisely those under which, according to Acemoglu and Restrepo (2018), the mismatch between the skill requirements of new technologies and the capabilities of the workforce is likely to constrain the productivity gains that technological diffusion would otherwise deliver. Moreover, theoretical contributions from the endogenous growth tradition — most notably Romer (1990) and Lucas (1988) — have established that sustained productivity growth depends not merely on the accumulation of physical capital, but on the effective incorporation of knowledge and human capital into the productive process. This insight makes the quality and adaptability of the labor force central to any analysis of technology-driven growth.

Against this backdrop, the present study addresses the following central question: to what extent did the adoption of digital technologies contribute to labor productivity growth in Brazil between 2010 and 2023, and which structural factors

conditioned that relationship? The general objective of this research is therefore to assess empirically the relationship between digital technology adoption and labor productivity dynamics in Brazil over the defined period, paying particular attention to sectoral heterogeneity and to the role of moderating factors such as workforce qualification, institutional quality, and digital infrastructure availability.

The academic and policy relevance of this inquiry is twofold. From a scientific perspective, the literature that applies frontier theoretical frameworks — particularly task-based models of automation and productivity — to the Brazilian case remains comparatively underdeveloped, especially for the post-2015 period, which combined severe economic contraction with accelerating digital transformation. From a policy standpoint, understanding the conditions under which technological investment translates into effective productivity gains is essential for designing industrial, educational, and innovation policies capable of placing Brazil on a more sustainable long-run growth trajectory. The social dimension is equally important: as Acemoglu and Restrepo (2018) emphasize, automation and digitalization reshape labor markets not only by displacing workers from routine tasks but also by creating new categories of employment that demand distinct skill sets — a transition whose management is critical in a country marked by deep labor-market inequalities.

This paper is organized as follows. Section 2 presents a review of the theoretical and empirical literature on digital technology adoption and labor productivity, with particular attention to the Brazilian evidence and the international comparative context. Section 3 describes the methodological approach, including the data sources, variables constructed, and the empirical strategy, which combines

sectoral panel data analysis with time-series econometric techniques. Section 4 presents and discusses the main results, highlighting sectoral heterogeneity in the productivity response to digital adoption. Finally, Section 5 summarizes the principal findings, acknowledges the study's limitations, and suggests directions for future research.